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Review

Critical review of electronic nose and tongue instruments prospects in pharmaceutical analysis

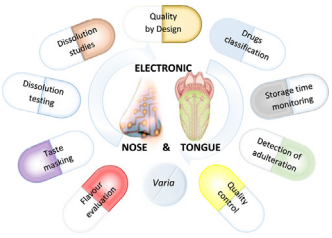
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HIGHLIGHTS

- Developments of electronic nose and tongue dedicated to applications in pharmaceutical analysis.
- Commercially available instruments.
- Concepts of bioelectronic type instruments.
- Critical discussion of current status and limitations.

GRAPHICAL ABSTRACT



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ABSTRACT

Electronic nose (enose, EN) and electronic tongue (etongue, ET) have been designed to simulate human senses of smell and taste in the best possible way. The signals acquired from a sensor array, combined with suitable data analysis system, are the basis for holistic analysis of samples. The efficiency of these instruments, regarding classification, discrimination, detection, monitoring and analytics of samples in different types of matrices, is utilized in many fields of science and industry, offering numerous practical applications. Popularity of both types of devices significantly increased during the last decade, mainly due to improvement of their sensitivity and selectivity. The electronic senses have been employed in pharmaceutical sciences for, among others, formulation development and quality assurance. This paper contains a review of some particular applications of EN and ET based instruments in pharmaceutical industry. In addition, development prospects and a critical summary of the state of art in the field were also surveyed.

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1. Introduction

Human senses of smell and taste are complicated and not fully understood systems. For instance, the olfactory system contains thousands of olfactory receptors (ORs) that bind odorous molecules, which allows detection at the ppt (parts per trillion) level or lower [1]. The structure of the olfactory system has been extensively studied and is currently well known. However, the mechanism of olfaction indicating which of the receptors and how discriminate thousands of odours, is still the subject of research [2]. In the case of other senses, the list of receptors is limited. Human eye has three receptor types, reacting to red, yellow and blue colours, and is capable of discrimination of a wide range of colour tones and shades [3]. In a canonical form, human tongue can identify 5 tastes using 28 different receptors [4,5]. Sweet, sour, bitter, salty and umami tastes can be sensed because of signal generation by 10,000 taste buds with 50–100 taste cells in each of them [6]. Recently, the taste of metallic and fat receptors was discovered (GPR40 and GPR1) [7].

The progress in the area of biomaterials engineering is determined by research on identification of interactions between the ligand and the OR [8]. Since Axel and Buck discovered the gene family encoding vertebrate ORs [9], the research in olfactory molecular mechanisms and signal transduction pathways has been improved. Additionally, many studies have been fulfilled to explain sense of smell principles [10–12]. According to Refs. [13,14] deorphanization of ORs is fundamental to unveil smelling sensory process; as to date only 10% of the human OR have known specificity and functional properties. ORs activity is influenced by specific physicochemical properties of odorant molecules and how this information is processed in the brain to produce an activation code that ultimately results in an olfactory perception [13].

Individual flavour perception is triggered by a binding event between substances and olfactory/taste receptors expressed on the surface of olfactory/taste neurons in the specific regions. Sensory quality of pharmaceutical products is connected with such properties as appearance, colour, odour and texture [15]. Proper design of these parameters and adjustment of the products to consumers' requirements call for participation and continuous supervision of the specialists with highly-developed sensory abilities. Training of the expert team is a complex, time-consuming and expensive process. Hence, automated methods are proposed, which are free from human error factors including oversaturation, subjectivism and fatigue. Application of the odour and taste sensors is believed to facilitate analysis of odorous and taste substances.

Changes in European [16] and other markets [17,18] regulatory requirements started the development of medicines intended for paediatric use and require the development of age-appropriate formulations. Particularly in children, flavour of a medication is important with respect to adherence and compliance. As a lot of active pharmaceutical ingredients exhibit an unpleasant feelings, taste masking has therefore become increasingly significant. Researchers focusing on paediatric drug development beforehand considered the use of etongues and enoses as supplementation, but they also needed more reliable data to use sensing systems as adequate tools in flavour assessment [19,20]. In order to put the application of ENs and ETs for pharmaceutical applications in the right perspective and to understand the inspiration of making them well-suited candidates for pharmaceutical applications, a short characteristic of operations and potentialities of these instruments will be presented. The application potential of electronic senses in drug-related industry is presented in Fig. 1.

There are several articles on the subject of EN and ET technology [6,21], including different applications [22–25], operation principles [26], sample preparation [27] and pattern recognition techniques as well as computational methods for data analysis [21,28,29]. In this review the authors extensively focused on application potential of electronic senses in pharmaceutical industry. A short introduction on fundamentals of the enoses and the etongues operation is followed by discussion of basic applications of these instruments in pharmacy. A special emphasis has been laid on critical evaluation of the usefulness of these instruments in drug-related fields as well as on presentation of prospects of application fields broadening.

2. Electronic nose and electronic tongue systems

Attempts to mimic biological senses with artificial olfactory systems are still under development. Their significant evolution during the recent years have been surveyed [30]. A concept of combinatorial utilization of a sensor array, reported by Persaud and Dodd [31], was fundamental for the construction of numerous electronic noses and electronic tongues. Generally, this can be based on the operation principle of biological sense of smell and the signal transduction mechanisms, in which a big number of olfactory and taste receptors react to the same chemical stimulus but with different affinity. The result is a complex pattern of neural responses correlated in the brain with the previously identified stimuli in order to provide recognition and discrimination [32]. An analogy in operation principles is illustrated in Fig. 2.

The mechanism of identification of odorous substances implemented in the electronic analogues closely resembles that present in human nose. The main element of the electronic noses is a sensor array. Stimulation of its sensory elements of particular types leads to generation of specific signals [21]. It is a suitable method of signal processing and data analysis, which enables generation of unique responses (odour fingerprints) for particular odorous samples [33,34]. Identification of particular gas samples is possible due to excitation of the sensors present in the electronic nose array but each sensor is excited to a different extent. The outcome is a characteristic set of responses from individual sensors. Interaction of the analytes with a receptor layer leads to physical and/or chemical changes resulting in signal generation. The type and properties of the sensors utilized in the array determine its

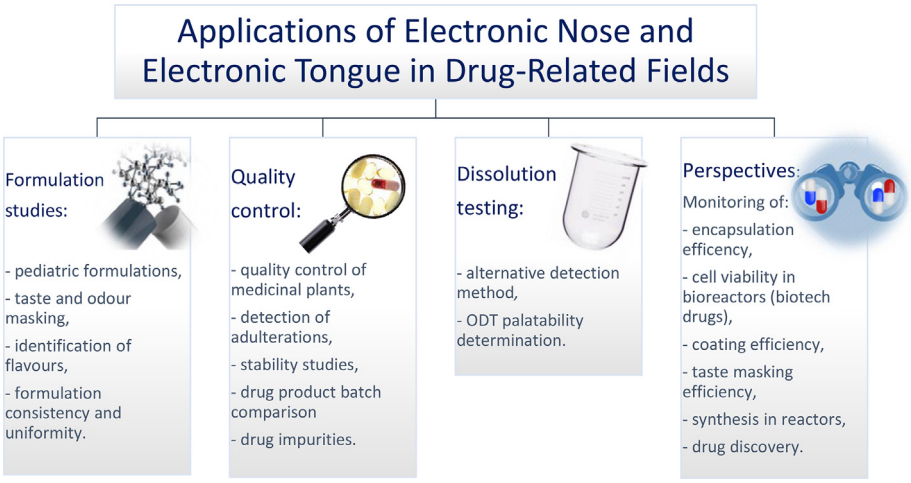


Fig. 1. Applications of enose and etongue in drug-related fields.

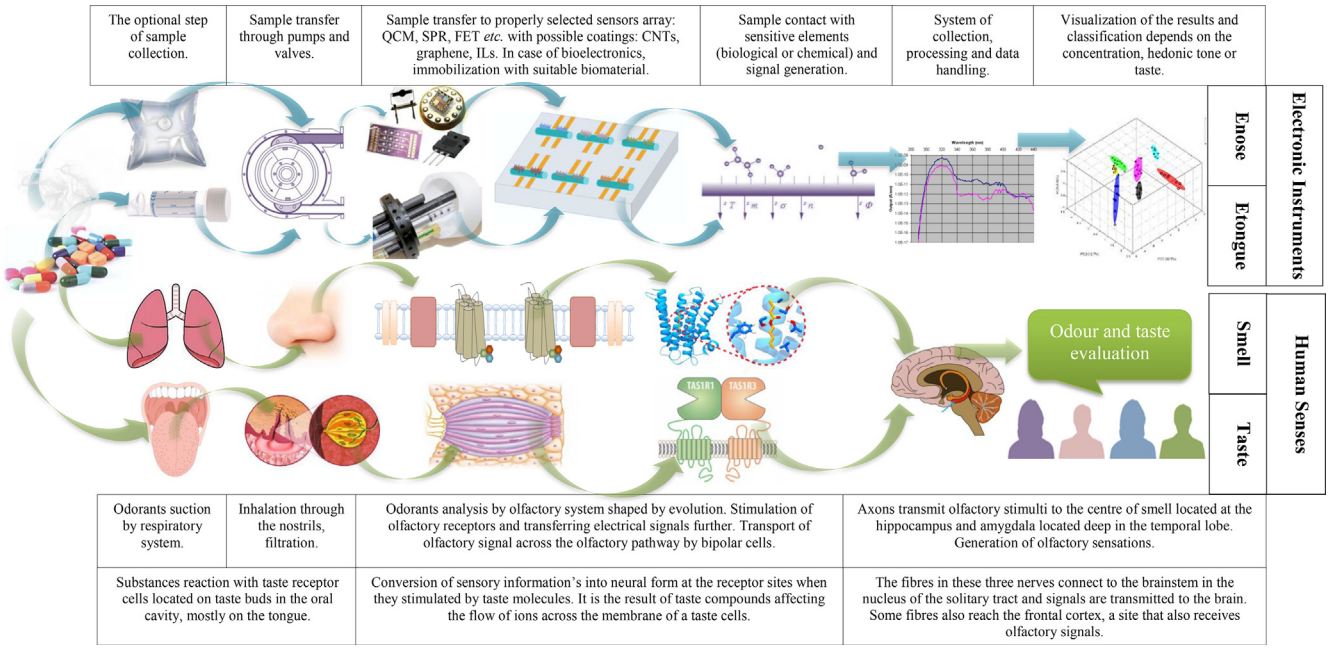


Fig. 2. Comparison of operation principle of electronic nose and electronic tongue; the analogy to biological systems.

analytical potentialities and the type of generated response as described in detail in the literature [35,36]. Analysis of the response vector allows classification of a given gas sample to the set of response vectors identified beforehand. In human, this role is played by brain which compares the nerve impulses from an olfactory bulb with the memory-stored odour base acquired during the lifetime. Unique features of discriminating the electronic nose devices from other devices are, among others, the possibility to employ chemometric data analysis and reducing time of analysis. The response vector is attributed to a particular set stored in a computer database using suitable data processing methods and statistic-chemometric algorithms [37,38]. Classification of gas sensors is usually based on signal transduction principle and the type of sensing material. Many different technologies of sensors [39] and biosensors [40] have been implemented so far.

Similar operation principle pertains to the electronic tongues. A system of sensors generates signals, which not necessarily have to be specific for particular compounds. The response is a set of signals, which can be correlated with particular qualitative properties of the sample. The applied sensors exhibit different cross-sensitivities to given substances dissolved in aqueous environment. Similar to the electronic nose, a suitable data analysis method is necessary to process the signal and classify the sample [28,41,42]. A range of sensors [43] and biosensors [6,44] that can be implemented in the etongues offers a wide spectrum of application, including food and pharmaceutical industry [45]. However, most of the commercial systems are based on potentiometric sensors, especially Ion-Selective Electrodes (ISEs).

Crucial factors during selection of commercial, sensor-based measurement systems are price, availability and specific character of target application. Many commercially available devices of EN and ET type have a wide range of application in different fields of practice and science. Table 1 shows the devices that can be utilized in pharmaceutical applications. Beside commercial systems, there

Table 1
Commercial electronic type instruments suitable for pharmaceutical analysis [47–49].

Manufacturer	Products	Sensor types	Application
Enose			
AIRSENSE airsense.com	PEN2, PEN3, I-PEN, i-PEN ET, iPEN3/MOD, OlfoSense	MOS	Flavour and fragrance testing, chemical analysis, environment safety
Alpha MOS alpha-mos.com	Fox*, Heracles, GEMINI*, Prometheus	QMB/CP, MOS, Flash GC	Quality control and storage, packaging evaluation;
Bloodhound Sensors/ Roboscientific roboscientific.com	ST214*, BH114*, 307 VOC Analyser	Conducting polymers	Quality control, flavour and fragrance testing, environmental monitoring, degradation detection
Cyrano Science Inc. cyranoscience.com	Cyranose 320	Conducting polymers	Quality control, freshness, spoilage and contamination detection
Electronic sensor Technology estcal.com	ZNose	SPR	Detect and analyse accurately organic, biological & chemical compounds in real time
FIS fisinc.co.jp	ODSA-P1, ODNA-P1,	Semiconductor gas sensors	Quality monitoring, environmental monitoring, health monitoring
GSG Analytical Instruments GmbH gsg-analytical.com	MOSES II*, VOCvario*, VOCcheck VOC meter*, VOC iDent*	Multiple sensor options	Classification of gas mixtures, aromas and flavours
Owlstone Nanotech, Inc. owlstonenanotech.com	Lonestar	FAIMS (Field Asymmetric Ion Mobility Spectrometry)	Quality control, purity control, identity control, freshness tests
Sacmi sacmi.com	EOS Ambiente/Aroma/ Portable	MOS	Quality control
Seacost science seacoastscience.com	SC-210	CP	Specific classes of chemicals
VOCscan Voscan.com	VOCscanner*	Mass spectrometry	Conformity and quality control, process optimisation and control, benchmarking in product development, correlations with sensory panels, medical diagnosis
3S Gas Sensing Solutions 3S-ing.de	3S-OdourChecker, 3S-SniffChecker, 3S-LeakChecker	–	Selective detection of gas mixtures
Etongue			
Alpha MOS alpha-mos.com	α-Astree, Astree II	ISFET	Quality control, taste assessment, process monitoring; company also offers electronic colour and shape visual analyser
Anritsu Corp. anritsu.com	Insent*	Potentiometric	Contaminant detection, quality control
Intelligent Sensor Technology Inc. insent.co.jp	Taste Sensing Systems SA402B*, TS-5000Z	Potentiometric	Quality control, enables detection of five basic tastes and also flavour attributes like “sharpness” and “richness
McScience Inc mcscience.com/	Multiarray Chemical Sensor*	PVC and polyurethane membranes	Quality control
Sensor Systems sensorsllc.com	Sensors	Potentiometric	–

MOS - Metal oxide semiconductor, CP - conducting polymer, QCM - Quartz Crystal Microbalance, SPR – Surface Plasmon Resonance, ISFET - Ion-Sensitive Field-Effect Transistor.
*Out of stock, available only on second-hand market.

is also the possibility to order taste assessment service. Aissy Inc., Tokyo, Japan, offers an accurate analysis of products by using its original taste sensor and compatible software, which can be helpful for the development of products, especially for researchers less familiar with sensors technology [46].

3. Applications in pharmaceutical sciences

The elaboration of new pharmaceutical products, their quality and usability control, especially if a drug contains easly-degradable substances, call for the analytical tools that are reliable and simple to use. Monitoring of the pharmaceutical products, with respect to quality and production processes control, is usually conducted using physico-chemical examinations, including pH and colour measurement, determination of the Active Pharmaceutical Ingredients (APIs) and excipients content, with the spectroscopic methods (Fourier transform infrared (FTIR), near infrared (NIR), ultraviolet–visible (UV–VIS) techniques, *etc.*). Evaluation of medicinal products with regard to their taste and smell is especially troublesome. In the majority of cases, application of the physico-chemical methods is not effective. On the other hand, practical problems associated with utilization of the sensory panels substantially restrict application of this method to routine investigations. Headspace gas chromatography (HGC) and two-dimensional gas chromatography with different detectors (for example mass spectrometry coupled with olfactometry) can provide detailed data about chemical composition as well as the information on organoleptic properties of particular volatile compounds present in a mixture [50]. Such analysis can be useful for elaboration of new products or detection of the components deteriorating organoleptic quality. However, they are very time-consuming and expensive. Despite the progress in instrumental analysis, taste sensations perceived by humans can be measured only with the sensory tests [51]. Volatiles and flavours can originate from various components of the pharmaceutical products [52]. They are present in raw materials used for drugs production but can also be generated during processing and storage. The taste is one of three main characteristics taken into account during foodstuff selection; appearance and texture are the other two. These properties are equally important for the selection of a drug by consumer. The sensations felt by a patient during drug application have a significant influence on the perception of effectiveness of the therapy [53], associated with certain parameters optimized at a drug formulation stage.

3.1. Masking the taste in the development of oral formulations

Taste is one of the main properties of oral drugs that may can have a negative impact on patient compliance. Most of the active substances and some of excipients are characterized by unpleasant, bitter taste [54]. Taste masking is particularly important in the case of paediatric and geriatric patients when unpleasant sensations

discourage the patients from further pharmacotherapy. That is why during development of oral formulations, especially the ones dedicated to these two groups of patients, it is necessary to apply a suitable taste masking technique. For instance, bitter taste of oral drugs can be overcome *via*: (i) addition of sweetening substances, (ii) complexing of the active substance, (iii) hot melt extrusion with polymers, (iv) coating the tablets with a polymer insoluble in oral cavity (for example Eudragit E) or (v) application of modern drug forms (microcapsules, microspheres, *etc.*). The routine methods for evaluation of oral drug taste masking are *in vivo* investigations conducted on adult, healthy volunteers using a suitable 'taste scale'. However, these tests provoke some ethical doubts. It is especially the case of substances at an early stage of development that do not possess a full toxicological profile. Moreover, these investigations cannot be employed for the paediatric formulations due to significant differences in physiology of taste between adults and children. Hence, new both direct (animal behavioural tests and cell-based assays) and indirect (consisting in evaluation of active substance dissolution) methods of taste masking assessment have gained increasing interest [19,55,56]. Another example of an alternative and direct method of taste masking evaluation can be utilization of the ET. So far, several papers describing application of the ET for the investigation of oral pharmaceuticals have been published. For instance, implementation of the ET in evaluation of taste masking effectiveness of ibuprofen and roxithromycin microcapsules [57], cetirizine hydrochloride microspheres [58], sodium benzoate paediatric minitablets [59], ebastine ODTs [60], clarithromycin oral suspensions [61], paediatric valaciclovir oral liquid preparations [62]. In all cases ET was able to differentiate products of different taste profiles. Additionally, for studies in which *in vivo* investigation was included, it was in good correlation with ET results. Interestingly, an unsuccessful attempt to apply ET analysis in praziquantel liquid formulation development was also described [63]. TS—5000Z (Insent Inc.) was not applicable for taste masking excipient evaluation, *i.e.* none of the ET sensors provided any significant response towards studied API. This state of affairs, as explained by authors, was due to non-ionic character and poor water solubility of praziquantel. In contrast, application of impedance spectroscopy-based ET allowed to evaluate taste masking capabilities of microparticulate formulations (microcapsules and microspheres) of praziquantel [64]. In this study an array of five sensors made by interdigitated gold electrodes coated with layer-by-layer films, obtained by alternating immersion into cationic and anionic solutions, was used.

3.2. Quality control of medicinal plants

An increased emphasis laid on the quality of pharmaceutical products also refers to medicinal plants – the drugs, which main components are raw herbal materials or herbal preparations. In spite of the fact that most of them have been known and applied for therapeutic purposes for hundreds of years, their proper evaluation with modern analytical techniques is indispensable. The medicinal plants are complex objects and every qualitative and quantitative deviation in the product composition may result in serious adverse effects. Moreover, year by year the patients become more interested in 'natural therapies'. Unfortunately, this also results in introduction of adulterated or low-quality products to the market. Chromatographic techniques (HPLC-UV, GC-FID), mass spectrometry coupled with chromatographic techniques (LC-MS, GC-MS), DNA analysis based techniques and microscopy techniques are routinely employed to quality control of the medicinal plants. All of these are characterized by a high complexity, long time of analysis (resulting from the necessity of sample preparation or/and time-consuming examination) and significant cost. Moreover, they do not provide real time analysis of the product. Unlike the aforementioned techniques, the EN is a simple, fast and direct analytical technique, which offers on-line testing of a product. Evaluation of the medicinal plant occurs *via* generation of an odour pattern characteristic of each sample. The pattern is created based on the signals from volatile metabolites collected by the EN sensors [65,66]. So far, many papers devoted to utilization of the EN instrument to evaluation of medicinal plants quality have been published. Most recent applications of EN to Quality control of medicinal plants are presented in Table 3. Generally, EN analysis additionally offered differentiation with respect to the country of origin or even to the particular crop origin. Because of that, it proved to be superior herb characterization technique when compared to HPTLC [67]. Moreover, EN in all cases was at least as efficient [68–70] or even superior [71] to GC-MS-based herb evaluation. This ability and significantly shorter time of analysis with the EN revealed a high application potential of the technique. On the other hand, it is interesting to note that unlike the *in vivo* taste tests, the EN analysis of *Lippia citrodora* infusions did not show any differences between samples prepared in different manner [72]. Another challenge in utilizing EN for herb quality control might be data processing. As an example, utilization of the average value of maximal response of particular sensors as a data set for statistical analysis was insufficient to discriminate the cultivation origin of *Angelica sinensis* [75]. Such discrimination was feasible when the ratios of the maximum readings from suitable sensors constituted the data set and when the EN results were analysed with suitable statistical methods (PCA, soft independent modelling of class analogies and hierarchical cluster analysis). Recently, the EN was utilized together with a number of analytical techniques, including GC-FID, GC/MS, UV/VIS and HPLC-UV-DAD, to characterize different populations of *H. officinalis* subsp. *Aristatus* [73]. The EN results were compatible with the ethereal oils and volatile organic compounds profiles; they allowed discrimination of a few main chemotypes of the herb. The results showed that the EN could either provide a supplementary instrument to routine analyses or else constitute the reference technique.

3.3. Dissolution testing

Dissolution is a phenomenon of transferring the active substance of a drug to the environment where it can be absorbed into organism. For solid forms of oral drugs, it is usually the first process occurring after drug application. Dissolution depends on qualitative and quantitative composition of the active ingredients and excipients as well as the parameters of the production process. That is why the investigation of the dissolution process is one of crucial methods of oral drug evaluation that found application in: (i) development of formulations, (ii) quality control of production batch, (iii) investigation of *in vitro – in vivo* correlation, (iv) investigation of bioequivalence of the formulations, for which there is no obligation to perform bioavailability tests. In practice, investigation of the dissolution consists in placing a drug in a respective medium, usually at 37 °C, at a volume exceeding 3–10 times the volume of a saturated solution of the active substance. Then changes in the therapeutic substance concentration are determined against time. UV-VIS spectroscopy and high-performance liquid chromatography with UV-VIS detection are the analytical techniques routinely used to measure the concentration [74]. The ET seems to be an interesting alternative in this field. This technique not only allows fast measurement in real time, but also provides the information about dissolution profile of the excipients. Such complex evaluation of a drug can be the source of additional information, for example about taste quality or it can support the investigations on *in vitro – in vivo* correlation of the dissolution process. Importantly, parameters

Table 2
Comparison of analytical techniques for measurement of drug concentration during dissolution testing [78,79].

Technique	Advantages	Disadvantages
UV/VIS	• Possibility of on-line analysis (flow device), • Cheap, easy to perform.	• Analysis limited to chromophore-containing compounds or species after derivatization by UV tagging, • Medium cannot contain the substance absorbing the light at the same wavelength as the analyte.
HPLC-UV	• Possibility of analysis of substances dissolved in complex media, • Possibility of simultaneous concentration analysis of few products, • Flow-through systems.	• Analysis limited to chromophore-containing compounds or species after derivatization by UV tagging, • More complicated than UV/VIS, • Requiring large quantities of expensive organics.
LC-MS	• Sensitive detection, determination of trace amounts • Simultaneous analysis of complex samples, • Analysis both chromophoric and non-chromophoric drug compounds, • Many ionization methods.	• Expensive equipment and running costs. • Adducts formations. • The need for a highly specialized personnel.
EN and ET	• Possibility of analysis of substances dissolved in complex media, • Possibility of analysis of substances with complex matrices, • Possibility of simultaneous concentration analysis of few products. • No limitation regarding chromophore-containing compounds, • On-line analysis.	• Complicated data interpretation, • Poorly recognized technique, • No guidelines how it can be used as a reference technique, • Time- and computation-consuming calibration procedures.

(medium temperature, pH and ionic strength) and conditions (stationary *vs* flow through) have significant influence on dissolution profile obtained using ET [75]. Based on study comparing ET-based dissolution profiles valsartan minitablets, elevated temperature led to more pronounced differences between the formulations. Moreover, utilization of artificial saliva was the most effective as far as discrimination of taste profiles of the minitablets was concerned. On the other hand, application of a phosphate buffer yielded a better consistence with the results of reference, pharmacopoeial method. Finally, the flow investigations were characterized by a higher repeatability than the stationary ones. Another important topic related to ET-based dissolution testing is mixture effect – a phenomenon occurring in the case of the drugs whose taste is masked *via* coating, for instance using a coating polymer that precludes tablet dissolution in oral cavity [76].The changes in taste profiles occurring during investigation of such formulations can result not only from the modified way of substance dissolution, but also from the interaction between particular components. Additionally, before investigation it is important to pay attention to the selectivity and sensitivity of the electrode with respect to particular analyte. As an example, unlike the reference method, the ET-based dissolution test indicated that the hypromellose-based granulate with pseudoephedrine sulphate was characterized by a more efficient release modification than metamizole granulate [77]. The authors explain this phenomenon in terms of a larger difference in electrode selectivity in the pseudoephedrine sulphate/hypromellose couple as compared to that of the metamizole/hypromellose couple. Overall, ET presents an interesting alternative to standard dissolution testing analytical methods such as, for example, UV/VIS. It would find application especially when: (i) the dissolution medium is too turbid; (ii) the analyte of interest carries no chromophores; (iii) the study involves simultaneous API and excipient concentration measurement. A comparison of the analytical techniques used for the measurement of the active substance concentration during dissolution testing is presented in Table 2.

3.4. *Varia*

Apart from the above-mentioned applications, the ET and EN instruments were employed in many other fields of pharmaceutical science, including quality control of solid, oral drug forms, quality control of aroma in oral solutions and adulterated drugs identification. In view of the wide range of the topics involved these applications were surveyed in the '*Varia*' chapter. As an example, the Fox 4000 EN was used for quality control of oral solutions with respect to applied flavour [80]. In another study, the authors evaluated scent-masking effect of different flavours (coffee, green tea, apple, fruit, plum and pineapple) on Aminoleban® EN – enteral nutrient product characterized by bitter taste and unpleasant odour [81]. In the next example, the αPrometheus EN was employed to odour analysis of the tablets with L-cysteine, and with different coating types that provided odour masking effect [82]. Lastly, the αFox 4000 EN and the flash-GC technique were applied to analyse adulteration (addition of phenolphthalein and/or sibutramine) of the slimming drugs [83]. Both techniques were capable of discrimination of the original and adulterated drugs. However, it was a combined statistical analysis of the data from both techniques that enables to distinguish the type of adulteration. Nevertheless, the EN turned out to be a perfect instrument for fast discrimination between the original and adulterated drugs.

The examples of electronic nose and tongue instruments applied in drug-related fields are set out in Table 3.

4. Future outlooks

Application of the electronic noses and tongues can be useful in pharmaceutical industry. Nevertheless, the electronic senses still possess some limitations regarding sensitivity and specificity as compared to their biological counterparts. Research on smell and taste fusion can play a key role in automation of evaluation, quality control and authenticity control of various products including pharmaceuticals. Another potential direction of the ENs and ETs development in pharmaceutical analysis are the bioelectronic noses (b-enoses, b-ENs) and bioelectronic tongues (b-etongues, b-ETs). They exhibit improved sensitivity and a broader response range than their electronic counterparts. Finally, the sensors characterized by sufficient repeatability and accuracy can be utilized as analytical instruments, monitoring production of therapeutic products within the process analytical technology (PAT). The aforementioned potentialities of sensors development are surveyed in more details in the following chapters.

Table 3
Applications of EN and ET in drug-related fields.

Application	Samples/Analytes/Dosage form			Instrument	Data analysis	Results	Ref.
Taste masking	Five samples for each API	Rixithromycin, Ibuprofen	Microcapsules	Warsaw laboratory system (eight sensors)	PCA	ET was able to distinguish samples containing APIs in free form and microencapsulated ones	[57]
	Six cetirizine formulations and three types of control samples	Cetirizine	Microspheres	Warsaw laboratory system (eight sensors)	PCA	ET provides information on both the amount of the dissolved API and taste of the sample	[58]
	Different sodium benzoate formulations	Sodium benzoate	Coated granules, granules, minitablets	TS5000Z αAstree	PCA	ETs: were able to detect differences between all formulations and sample API, -confirmed the results of the dissolution test, -were able to detect differences between all granules and water	[59]
	Ten samples of ebastine-loaded orally disintegrating tablets (ODT)	Ebastine	ODT	αAstree	PCA	ET was able to evaluate the bitterness of ebastine ODTs using the Euclidean distance	[60]
	Three samples of CM: Klacid and two generics	Clarithromycin	Suspension	αAstree	PCA	ET was capable for taste assessment of oral suspensions and predicting volunteer responses	[61]
	Three formulations in two different concentrations	Valaciclovir	Liquid	Insent taste sensing system SA402B	PCA	ET was able to single out the formulation characterized by superior palatability characteristics for further *in vivo* evaluation.	[62]
	Twenty one sample types	Diclofenac	Liquid	Warsaw laboratory system (eight sensors)	PCA	PCA analysis revealed that the masking efficacy was altered by the structure of the sweetener addition.	[84]
	Five model drugs with four common taste-masking sweeteners	Paracetamol, Ibuprofen, Tramadol hydrochloride, Sildenafil citrate	Liquid	αAstree II	PCA	ET was able to evaluate taste masking effect against common sweeteners. Sensors responded with good selectivity and sensitivity (10 mV)	[85]
	Samples consisting of various concentrations of famotidine, sucrose and sugar alcohols	Sweeteners	Liquid	Insent taste sensing system SA402B	Univariate	ET was able to predict directly the bitterness intensity of the mixed solution and evaluate the bitterness intensity of the dissolution media of commercially avail- able ODTs	[86]
	Six samples: one API and five formulations	Acesulfame K, aspartame, citric acid	Sublingual tablet	αAstree	Multivariate (PLS – Bitterness prediction module)	ET was able to screen different non-medicinal ingredients to determine the agent that best masks the unpleasant taste of the API, especially at the early stages	[87]
	Nine different ODTs: eight generic products	Famotidine, Aspartame	ODT	αAstree	Multivariate (PCA – Euclidean distances)	ET was able to evaluate the bitterness-suppressing effect and to predict the bitterness	[88]
	Fifteen bottled nutritive drinks	Thiamine, thiamine derivates, medicinal plant ingredients	Liquid	αAstree and FOX-4000	Univariate and multivariate (PCA)	The sourness and bitterness intensity was predicted by instruments. Overall palatability is positively correlated with sourness intensity and fruitiness ($R > 0.82$) and negatively correlated with bitterness intensity and the tasting of medicinal plants ($R < -0.80$).	[89]
	Four samples	Sildenafil citrate	Microcapsules	αAstree II	PCA, DFA	ET was able to precise evaluate taste masking efficiency with excellent correlation to *in vivo* sensory tests	[90]
	Seven samples	Cholorpheniramine Tannate complexes	Powder	αAstree	PCA, DFA	ET was able to evaluate taste masking efficiency and predict best ratios in drug formulations	[91]
	One aqueous solution	Praziquantel	Liquid	TS5000Z	N/A	ET was not able to obtain concentration-dependent signal for API in the full concentration range	[63]
	Six formulations with their respective placebos and constituent mixtures	Praziquantel	Microparticulate formulations	Impedance spectroscopy-based ET	IDMAP, SAMMON, PCA	ET was able to evaluate taste masking efficiency of studied formulations	[64]
	Eight formulations	Potassium Chloride	Syrup	TS5000Z	N/A	Dry syrup containing KCl coated with 10% Eudragit E 100 solution could effectively mask unpleasant taste of the drug	[92]
	Seven formulations and fourteen API/excipient mixtures	Diclofenac	Fast-dissolving film	Insent taste sensing system SA402B	PCA	ET analysis was in agreement with human test panel	[93]
	Six formulations	Cetirizine dihydrochloride	Microparticles	Warsaw laboratory system (eight sensors)	PCA	ET, in line with release test and human taste panel, was able to evaluate taste masking effectiveness	[94]
Quality control				MOSES II	PCA		[67]

Table 3 (*continued*)

Application	Samples/Analytes/Dosage form			Instrument	Data analysis	Results	Ref.
	Nine samples differing in species, origin and supplier	*Valeriana officinalis*, *Valeriana wallichii*				EN discriminated samples based on their species and origin. Additionally, results were comparable with those obtained by TLC analysis.	
	Seven samples differing in drying method and storage time	*Lippia citrodora*		EOS 835	PCA	EN discriminated samples based on their storage period.	[72]
	Six samples (fresh and dried herbs)	*Rosmarinus officinalis*, *Ocimum basilicum*, *Salvia officinalis*		10 MHz QCM with molecularly imprinted polymers	–	EN discriminated fresh samples from the dried ones, which was in line with GC-MS analysis.	[68]
	Five samples of fresh herb differing in Italy region and four samples of dried herb differing in country of origin	*Glycyrrhiza glabra*		ISE Nose 2000	DFA	EN enabled effective identification of the products with respect to the degree of its processing (fresh vs dry material) or crop origin.	[69]
	Samples differing in age and planting bases	*Panax ginseng*		FOX-4000	PCA, DFA, HCA	Application of backpropagation algorithms of neural networks to evaluation of the root age provided a better accuracy as compared to that of the GC/MS analysis	[71]
	Two different *Ginseng* species	*Panax quinquefolius*, *Panax ginseng*		FOX-4000	PCA, PLS	EN can be used not only to discriminate between ginseng species but also to predict quantitative content of the aroma components.	[70]
	Twenty-one samples differing in growing location	*Angelica sinensis*		FOX-3000	PCA, SIMCA, HCA	Sample origin discrimination was feasible when the ratios of the maximum readings from suitable sensors constituted the data set and when the EN results were analysed with suitable statistical methods	[95]
	Twelve samples differing in growing location	*Angelica sinensis*		FOX-4000	PCA, DFA	EN was able to successfully identify *Angelica sinensis* samples from different producing areas	[96]
	Samples differing in storage time	*Lonicera japonica*		FOX-3000	PCA	EN was able to successfully differentiate *Lonicera japonica* samples based on storing time	[97]
	Five samples differing in growing location	*Hyssopus officinalis*		PEN3	PCA	EN provided a significant supplementary instrument to routine analyses in herb characterization	[73]
	Eight different plant samples from *Asteraceae family*	Different plants from *Asteraceae family*		FOX-3000	PCA, PLS	EN was able to differentiate *Asteraceae* plants	[98]
	Thirty eight samples	*Cynomorium songaricum* from different growing areas in China		αAstree II	PCA, LDA	ET was able to classify the *C. songaricum* with 100% degree of accuracy	[99]
	Three different ibuprofen formulations	Ibuprofen-based formulations		Three different ETs	PLS-DA	Flow injection analysis coupled with ET exhibited higher efficiency of differentiation pure and modified ibuprofen samples when compared to classical method	[100]
Dissolution testing	Coated and uncoated minitablets	Valsartan	Minitablets	Warsaw laboratory system (eight sensors)	PCA	Dissolution parameters (medium temperature, pH and ionic strength) and conditions (stationary vs flow through) have significant influence on dissolution profile obtained using ET	[75]
	Eighteen API/excipient mixtures	Metamizole sodium or pseudoephedrine sulphate	Liquid	Warsaw laboratory system (eight sensors)	PCA	Chemical profile of the mixture was subjected to changes depending on a ratio of the components.	[76]
	Ten API, API/excipient, formulation samples	Metamizole sodium or pseudoephedrine sulphate	Liquid/ Grantulate	Warsaw laboratory system (eight sensors)	PCA	The majority of acquired dissolution profiles were consistent with those obtained with the pharmacopoeia method. However, unlike the reference method, the ET approach revealed that the granulate with pseudoephedrine sulphate was characterized by a more complex dissolution profile. Thus, before investigation it is important to pay attention to the selectivity and sensitivity of the electrode with respect to particular analyte.	[77]
	Two model tablets	Ibuprofen or quinine	Tablet	potentiometric multisensor system	PLS	ET, as opposed to traditional UV–Vis dissolution testing, was able to generate dissolution profiles. The dissolution profiles differed depending on applied medium.	[101]

(*continued on next page*)

Table 3 (*continued*)

Application	Samples/Analytes/Dosage form			Instrument	Data analysis	Results	Ref.
Drugs classification	Five samples (cannabis buds, cannabis plants, hashish, snuff tobacco, tobacco leaves)	*Cannabis sativa*		Portable electronic nose prototype	PCA, MANOVA, SVMs	EN was able to successfully differentiate illegal drugs	[102]
	Ten different plant samples from *Zingiberaceae* family	*Zingiberaceae* family		FOX-3000	Feature extraction and cascade classifier	EN was able to differentiate *Asteraceae* plants	[103]
Flavour evaluation	Eleven samples differing in flavouring agent, flavouring agent lot and storage time	Six flavouring agents	Liquid	FOX-4000	PCA, DFA, PLS	EN was able to assay the flavour concentration and monitor stability during shelf-life	[80]
	Eleven tablet samples differing in coating method	L-cysteine	Tablet	αPrometheus	PCA	EN was able to assess the odour-masking ability of tablet coating	[82]
	Seven samples differing in added flavoured powder	Aminoleban® EN	Liquid	FOX-4000	Multivariate (PCA – Euclidean distances)	ET was able to detect odours below the human threshold of detection	[81]
	Different liquids	Milk, orange juice, NaCl, sucrose, quinine, ethanol, ethyl acetate	Liquid	ET (SH-SAW based), EN (chemFET based)	PCA	Only combination of ET and EN was able to discriminate all studied samples	[104]
Drug adulteration	Four capsule formulations with illegal additives (3) and without illegal additives (1)	Phenolphthalein, sibutramine	Capsules	FOX-4000	PCA, LDA	ET was able to discriminate samples without illegal additives from the ones with illegal additives	[83]

4.1. Process analytical technology

The Quality by Design (QbD) concept implemented in pharmaceutical industry assumes that the quality of a product cannot be verified only at the end of production cycle but it should be incorporated in the entire process. Moreover, it puts an emphasis on a thorough understanding of the product, based on scientific fundamentals, investigations and risk analysis. One of the main aspects of QbD is the process analytical technology – a system aimed at control and better understanding of a production process. In practice, it consists in, among others, equipping the process with modern analytical techniques analysing Critical Quality Attributes (CQAs) of the products in real time. Such approach significantly minimizes the losses caused by disqualification of faulty products, because it enables an immediate correction of all abnormalities. The ETs and ENs seem promising candidates for the analytical techniques in PAT owing to their real time generation of results and non-destructive analysis. For example, the EN can be used for monitoring of: (i) encapsulation efficiency of volatile organic compounds, (ii) coating efficiency of the tablets with an active ingredient characterized by unpleasant odour (iii) cell viability in bioreactors (biotechnological drugs). Furthermore, the ET could be applied in monitoring of: (i) masking efficiency of bitter taste of oral solutions (ii) chemical synthesis in reactors [105]. On the other hand, ENs and ETs are still at their infancy regarding their applicability at regulated laboratory environments. Among others, they experience difficulties in reproducibility [106] and sensitivity [107]. Thus, further studies regarding developing a unified system preparation and operation approaches (*e.g.* universal external standard for data normalization) are mandatory. A significant milestone was achieved with the successful performance qualification of ET according to ICH Q2 [108].

4.2. Bioelectronic nose and bioelectronic tongue

In some aspects the bioelectronic nose differs from conventional electronic noses [109] where the active material is made of chemically sensitive elements with a broad response range. A principal assumption in design of the bioelectronic noses is application of the proteins of olfactory receptors as active elements of the sensor for odour analysis with high sensitivity and specificity. The sensory part of the bioelectronic nose can be built of the ORs or of the cells exhibiting expression of the olfactory receptors proteins. The b-EN, as one of the variants of electronic nose, is characterized by a high sensitivity owing to application of a biological receptor element on the surface of the secondary transducer. It enables an identification of particular odorants regardless of physical properties and complexity of the sample matrix.

The ORs used in the b-ENs belong to the group of G-protein-coupled receptors (GPCRs), which is a family of integral membrane proteins possessing seven transmembrane helices [110]. GPCRs are engaged in important physiological processes, including sensory signalling, neuron transmission and hormonal signalling [111]. As a result, more than a half of the currently available drugs are GPCRs control oriented [112]. The concepts similar to the bioelectronics noses can be employed to elaborate a GPCRs functionalized sensor, for example to biomedical and pharmaceutical applications – diseases diagnostics or drug discovery. Some neurotransmitters such as dopamine, serotonin, glutamate and acetylcholine play a key role in the operation of the central nervous system and in many neurological disorders. They can be potentially monitored with the GPCRs functionalized sensor [113]. The GPCR-based sensors can also be used in drug discovery and in basic physiological studies on GPCRs including de-orphaning, antagonism, and dimerization [114,115]. The main limitation in the development of these sensors is GPCRs production from different types of cell systems. Due to complicated structures and different mechanisms of GPCR, their overexpression, solubilisation and purification are limited [116]. A progress in protein expression methods and purification using bacterial cells cultures enables GPCRs large-scale production [40]. Thus, it is expected that different types of GPCRs functionalized sensors implemented in the bioelectronics noses are expected to offer a significant application potential in biomedical and pharmaceutical fields.

A concept of the bioelectronic tongue for taste evaluation in pharmacy is increasingly popular in the literature [117]. These devices operate in the analogous way to the b-ENs, the receptor element of the sensor being of biological origin, such as enzymes, peptides, antigens, antibodies, receptor proteins, cells and tissues. Although the b-ETs have already been employed in taste evaluation

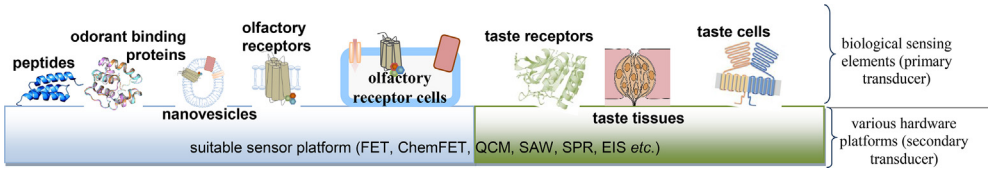


Fig. 3. Scheme of bioelectronic noses and tongues coated with various biological materials.

and selective detection of the components of different products, it should be emphasized that they are at an early development stage, similar to the b-EN instruments. The artificial nose and tongue devices may greatly improve functionality and efficiency of sensors for smell/flavour detection that mimic natural smell and taste allowing elucidation of the biological parts [118,119]. Apart from typical biological components of the olfactory systems [120], odour biosensors utilize the structures that mimic biological materials such as synthetic peptides [121–124]. The structure and characteristics of these materials are designed to act as binding sites in the olfactory receptors. The key issues in biosensors production are the type of biosensing element as well as the technique of immobilization, which have a direct impact on metrological parameters of the biosensor [125].

The electronic tongue instruments can generate analytical data useful in the analysis of multi-component matrices. However, there is still a gap between the ET systems and biological taste, caused by receptors structure and coding mechanism. In this case, a reasonable solution can be provided using biosensors, whose structure is based on the materials of biological origin allowing highly selective and sensitive analyte detection [126]. To achieve taste transduction, a response from the cells of taste receptor is first coded in taste buds of the epithelium *via* a change of action potentials. Accordingly, application of the taste buds as receptor materials should result in high effectiveness of taste identification by the bioelectronic tongue. Liu et al. implemented a microelectrodes arrays (MEAs) to record electrophysiological activity of taste epithelium, which allowed identification of the fundamental tastes [127] (Fig. 3). Similar investigations were conducted to identify salty [128] and umami [129] tastes. Moreover, a 60-chanel planar MEA device was used to analyse the sweetening substances; a mixture of glucose and saccharose, saccharin and cyclamate was utilized as the reference sample [130]. Wang et al. [131] presented a biosensor, whose sensitive element was a membrane fraction of HEK-293 cells containing human bitter receptor T2R4, extracted and deposited on the QCM sensor. The biosensor exhibited high specificity and sensitivity (1.21 kHz mM^{-1}) to denatonium. A concept of cell-based b-ET was developed by Qin et al. [132] for the high-specificity isothiocyanate-induced bitter detection, utilizing human Caco-2 cells as a primary sensing element and interdigitated impedance sensor as a secondary transducer. This study suggests that the constructed b-ET is a promising tool for specific detection of N–C=S-containing compounds that can be used, among others, in drug development and pharmaceutical analysis-based studies. A high-performance bioelectronic tongue with ligand binding domain T1R1 VFT for umami taste detection, with a high potential in pharmaceutical field, was developed by Ahn et al. [133]. The biosensor offered real time detection of L-glutamate (MSG) at a sensitivity level of *ca.* 1 nM. Umami taste intensification was amplified by monophosphate (IMP). In addition, the bET allowed efficient reusable property and storage stability. Thanks to the significant progress in biotechnology and nanotechnology, a combination of biological elements and nanomaterials for selective identification of compounds contributes to the development of nanobiosensors. Most of the biological processes involve electrostatic interactions and charge transfer, which can be determined directly using an electronic device based on nanomaterials [134]. Promising investigation fields of odour biosensors are connected with miniaturization such as the micro-flow systems. An attempt to integrate the bioelectronic nose with the micro-flow system (μBN) for trimethylamine (TMA) identification was reported by Park et al. [135]. Also, great efforts have been devoted to implementation of nanomaterials for olfactory-based biosensors. Biological signals can be transformed into electrical potentials by nanomaterial-based transistors [136]. These biosensors ensure a high selectivity of natural systems and a high sensitivity of nanomaterials. The bioelectronic sensors based on nanomaterials can provide perceptional codes and can be used in different practical applications, including drug-related fields [126]. Many nanomaterials with the dimensions adjusted to deposition of biomolecules, such as carbon nanotubes (CNTs), graphene, conducting polymer nanotubes (CPNT) and silicon nanowires (SiNW) were applied in the biosensors owing to their unique physico-chemical properties, which substantially improved sensitivity and selectivity [137]. Bioelectronic noses and tongues are expected to be used as next-generation analytical tools for rapid on-site quality and safety monitoring [126]. These concepts are still at an early stage of development but further advances in practical applications in pharmaceutical analysis can be expected. Significant challenges need to be met so that biosensors can be applied in drug-related fields [138]. Presently, gas biosensors still have several limitations. The successful commercial application and utilization in routine analytical practice needs to overcome the limitations imposed by durability, stability and repeatability [6,25]. Improvement in stability in aqueous media, control of which is mostly needed. Furthermore, future advances in biosensors will require integration of microfluidics and nanocomponents, reusability, portability for onsite analysis, reduction of costs and assay time. While this is a new, prospective and rapidly growing area of research, it is still at a stage of biosensor development.

Fig. 3 presents exemplary devices from the bioelectronic senses group.

4.3. Data fusion

Although the smell and taste are separate senses, they are interconnected and in the majority of cases there is an interaction between the two to provide perception of flavour. Moreover, the smell and taste senses are functionally linked to the digestive system [139]. Evidently, the temporary deterioration of the correct performance of the sense of smell such as overproduction of mucus in the cold, results in changes of the perceived taste of meals. Other also affect the perception of flavour, generating a significant flow of information to be processed by brain.

A set of data from different sensors makes up a complete image of a sample. Integration of the information obtained using various analytical tools is termed ‘data fusion’ and it encompasses the processing methods of multivariate signal of different origin [140]. Among the techniques employed for evaluation of food and pharmaceutical products [141], the last decade also witnessed a

combination of the electronic nose and tongue as two fast, non-invasive, non-destructive and reliable analytical instruments [142]. Fusion of the electronic nose and tongue can substantially improve classification of samples, especially when variations in the composition of the gas and liquid phase are vital.

The limitations of combined techniques utilization arise mainly from the fact that both systems are operated separately as experimental conditions for the EN and the ET activities are different. This can be the source of errors during data interpretation. There have been some attempts to construct coupled instruments [37,104,143]. It is also expected that combination of the EN and ET with an electronic eye module or computer vision can contribute to improvement of the samples classification system [144–146]. The attempts of coupling two or three systems performed so far confirm that the fusion results are more reliable than the ones obtained from particular systems separately and can offer more complementary information's [147]. Also, more improved feature extraction techniques can be developed for electronic sensor response analysis. Instead of using a unified sensor, development of an application-specific sensor would result in a better quality of analysis and this field is considered as another research direction. It can be concluded that combination of different detection systems brings many benefits to sensory analysis. However, the combined systems are still in a development phase and they require more investigation to be successfully utilized in pharmaceutical sciences. A combined ET/EN flavour system comprising of the SH-SAW liquid sensors and the chemFET gas sensors capable to discriminate between different flavours with excellent accuracy has been developed by Cole et al. [104].

The use of fusion technology between the electronic nose and the electronic tongue is displayed in Fig. 4.

4.4. Challenges

ENs and ETs, including both commercially available and laboratory prototypes, may show dissimilar performances with a standardised sampling and analysis protocols [148,149]. The stability of sensors and enhanced cross-sensitivity is a critical factor, which is defined as reproducible response of a sensor to as many species in solution/vapor as possible. If properly designed and trained (*i.e.* in terms of machine learning), the EN/ET have the potential to determine quantitative and, in some extent, qualitative composition of different samples [47]. However, there are certain issues in ETs/ENs instruments that preclude wide spread in routine analytical practice. An important obstacle relates to drift effects, observed mainly as a result of gradual change of sensor characteristics occurring during sensor exploitation and variations of environmental conditions. Drift effects may hinder the repeatability of analysis in different experimental sessions, endangering long-standing instrumental training and preventing from building-up of long-term volatiles/liquids databases [150]. Additionally, relative humidity might substantially affect composition of the gas phase in both quantitative and qualitative manner [151]. Another common issue are matrix effects (interferences) of analysed samples that could potentially alter sensor responses. The background interferences caused by cross-sensitivity and ambient factors can significantly decrease performance of ENs and ETs. Therefore, it is crucial to study interference suppression techniques. Since the interferences of ENs and ETs are unknown and unstable, some nonlinear methods have good effects for interference suppression, such as those based on transfer learning, adaptive methods, *etc.* [152].

Inter-batch homogeneity in ET/EN sensor production is one of the key quality attributes. The homogeneity entirely depends on the sensors embedded in ENs/ETs. Generally, the pattern recognition module used internally is the same among ENs/ETs and different inputs from sensors would lead to different signals. Unfortunately, sensors of identical types used in ENs/ETs sensor arrays reflect diverse signal responses to the same compounds in the same experimental conditions. This is mainly due to the integral sensors variability and discreteness in the manufacturing processes [153,154].

Specific correction methods have already been developed to solve problem of sensors discreetness [155,156]. Also, a variety of algorithms have been proposed to overcome the sensors drift problems [157,158]. Alternative approach for extension of sensors calibration lifetime can be achieved through mathematical drift corrections. A single sensor standardization approach (similar to single wavelength standardization recommended in spectroscopy) can significantly improve the extent of calibration lifetime [159]. Sensor responses are automatically normalized after instrumental calibration in several modern commercial ENs and ETs, thus compensating the systematic error in the output. In such conditions, instruments may achieve a remarkable stability even after

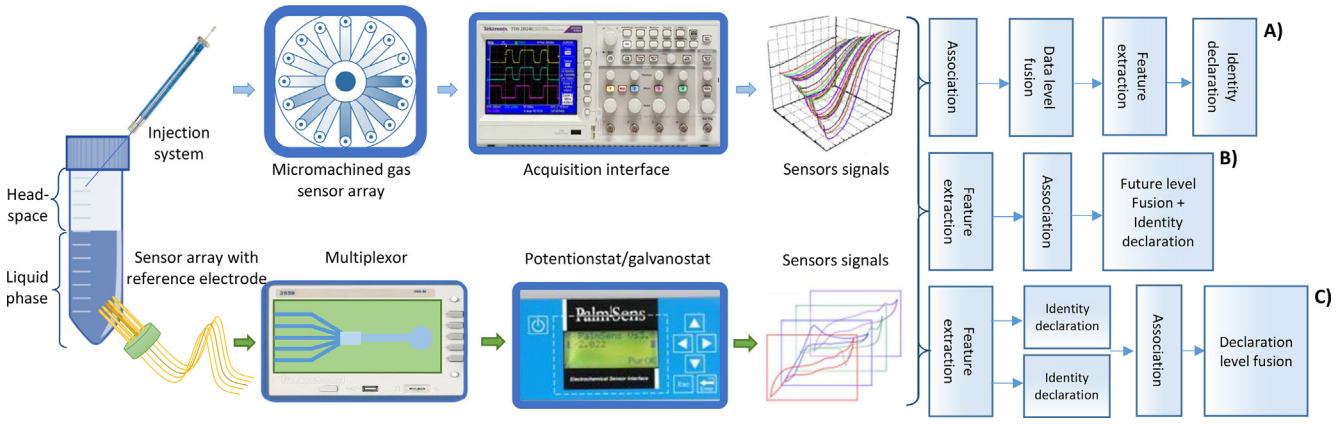


Fig. 4. The scheme of the fusion between the electronic nose and the electronic tongue with different data fusion methodologies [140]. A) Low fusion level: data from all sources are simply concatenated sample-wise into a single matrix that has as many rows as samples and as many columns as signals measured by the different instruments. In order to fuse raw sensor data, the original sensor data must be commensurate. B) Intermediate fusion level: some relevant features are first extracted from each data source separately and then concatenated into a single array that is used for multivariate classification and regression. Typically scores from PCA or PLS-DA are used. C) High fusion level: a separate model is built for each available data set, and then, the model responses are combined to produce a final "fused" response. Usually used for sample classification; each of the available models predicts a class label, and the class label that is predicted by the majority of the models is selected.

years [160]. However, most of the proposed solutions are linked with high computational costs *via* time-consuming recalibrations. Algorithm-based recalibration procedures can significantly increase instrument-feasibility and long-term performance assessments, as well as instrument design and the validation process [161,162]. As an alternative to the recalibration of the sensor systems using full set of calibration samples, calibration update and drift correction for ENs and ETs have been proposed [163].

5. Conclusions

The electronic noses and tongues are increasingly implemented in combination or instead of the traditional techniques of sensory analysis, including sensory panels, spectroscopic sensory analyses, *etc*. They outperform the classic techniques as far as the objectivity, repeatability and durability are concerned. Their fundamental qualities are mainly utilized in the fields where the offered products should be homogenous and subjected to routine, standardised tests, which should be fast, easy to execute and provide information on the quality of the product. Generally, application of the electronic senses in pharmaceutical industry should be more convenient than in foodstuff analysis because the pharmaceuticals are characterized by less complicated flavour profiles and matrices. Moreover, data analysis methods are being developed and applied to electronic type instruments, to combine characteristic responses and receive holistic image of analysed sample and better replicate human senses. This facilitates calibration and does not restrict the analysis to particular substances. Utilization of the electronic senses in analytical practice of pharmaceutical products, for instance in elaboration of drugs, quality control, taste masking, dissolution testing, is gradually becoming a real perspective. The instruments have satisfactory sensitivity and selectivity to perform flavour analysis in pharmaceutical products. The ENs and ETs are able to distinguish the differences between freshly prepared and aged samples. Therefore, instruments can be used for testing of flavours raw materials and flavoured solution formulations. Also, electronic type instruments can be used to evaluate the flavour concentration during release testing, to support packaging selection and to monitor flavour stability during shelf-life.

In the recent years, ENs and ETs have shown an important improvement in many aspects, *e.g.* decrease in time response and costs. More specifically, the latest works describing the electronic type systems for detection of drug ingredients demonstrated substantial progress in development of high sensitive sensor arrays based on different methods of detection and the use of enhanced data analyses. Although all applications presented in this paper seem very promising, the EN and ET instruments are still at the early development stage, especially the bioelectronics senses. Taking these into consideration, many research teams must contribute to improve usefulness of these devices in pharmacy. Compensation of signal drift caused by sensor ageing and surface contamination, high limits of detection, long regeneration time, low repeatability are just some points encountered in particular sensor groups to be solved. Hence, there is a need for optimisation of the sensors production and operation parameters aimed at improving accuracy and reliability. As a result, several methods have been already proposed to tackle the issues of temporarily sensor drift and calibration transfer between instruments and matrix effects. Another obstacles hindering the widespread of ENs/ETs is the limited temporal validity of the multivariate calibration models. Frequent recalibration of sensors arrays is often both time and money consuming due to the large number of necessary reference samples and their limited availability. Thus, the time- and computation-consuming calibration procedures are the key limiting factors of the widespread use of ENs/ETs. Nevertheless, the EN and ET technology has expressively grown since the first efforts of applications to pharmaceutical analysis and is likely to become routinely used. Further development of the ENs and ETs is driven by the emergence of novel types of sensors. Another challenge is the miniaturization of ENs and ETs including integrating them with biomaterials and nanoelectronics.

The value of their related research and development is seriously underestimated and needs to be re-evaluated. Precise and fast analysis, which does not require highly-qualified personnel, makes the ENs and the ETs a promising alternative to the time-consuming and expensive classic analytical techniques. In our opinion, the interest in the electronic senses in pharmaceutical applications will be still growing, thus enabling implementation of commercial devices into analytical practice.

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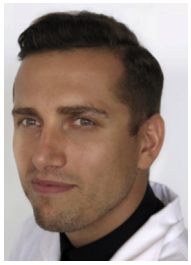
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